

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've been writing software for about 10 years, with the last three as a Software Architect on an enterprise insurance platform. Most of my work is backend and API: I own four REST APIs that internal teams and external partners build against, so a lot of my time goes to contracts, versioning, and the design standards that keep them stable as they grow. I care about the things that make an application hold up in production: scalability, reliability, and tests you can actually trust. OO design and the usual design patterns are how I structure a system, I lean on Postgres for data, and I'm comfortable in AWS and Kubernetes along with the CI/CD around them. I like picking up unfamiliar tools, and I like leaving systems more reliable than I found them.

Work Experience

Alberta Motor Association

SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Own 4 REST APIs serving 5 internal teams and external partners. Designed the contracts and versioning, approve new endpoint additions, and maintain OpenAPI design standards in Confluence. (Python, REST, OpenAPI, TypeScript)
- Led a 2-3 engineer rewrite of a legacy document generation system into containerized microservices on EKS. 4-5 services (with sidecars) on a 3-5 node cluster, deployed with Helm. Wired up Datadog metrics for failure rate, throughput, generation time, and storage time, replacing a system that had frequent outages. (Kubernetes, EKS, Docker, Helm, Datadog, C#)
- Set technical standards for a 12-person team through code reviews, RFC authorship, and PR checklists / API design guides. Member of the Architecture Steering Committee. Help newer developers navigate legacy codebases and trained the team on TDD methodology.
- Architected the infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, 99.9% uptime since launch, 2.8M insured trips and 76,637 invoices processed. Won the 2024 Guidewire Innovation Award (1 of 3 worldwide). (Python, AWS Lambda, Terraform, TypeScript)
- Built a self-service developer platform solo that took environment provisioning from 2-3 weeks down to 2 hours (60x). Self-service create/destroy, database snapshots, and pull-from-prod, adopted by 15 people across 3 teams and used to spin up 120+ environments. It made production debugging routine: a cancellation bug nobody could reproduce got root-caused in days using snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Built an event-driven pipeline that ingests an external partner's policy data: S3 ingestion, a Kubernetes CronJob, a translation service, and transactional calls to a composite API, with OAuth 2.0 auth and Datadog monitoring. (Python, Kubernetes, AWS S3, OAuth 2.0)

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and maintained backend services that ran medical AI algorithms and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team (15 developers / all Guidewire projects) and still in use 4+ years later. (Gosu/Java)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

Lockheed Martin Canada

ASSOCIATE SOFTWARE ENGINEER

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

The Open University MSc. COMPUTING (SOFTWARE ENGINEERING) Currently pursuing Master's in Computing at The Open University.	Milton Keynes, England (Remote) May 2024 - Ongoing
The King's University B.S. IN COMPUTER SCIENCE	Edmonton, Alberta Sep 2012 - Dec 2016

Skills

Backend & API Design	I own four REST APIs that internal teams and external partners build against, so most of my attention goes to the contract: clean versioning, sane error handling, and design standards that hold up as the surface area grows. I came up on the JVM and still reach for OO design and the standard patterns when I'm structuring a service. Designing backends that other people depend on, and not surprising them, is the part of the job I care most about.
Quality & Release Engineering	Earlier in my career I designed an automated test framework that cut defects by 30% on later projects; the insurance engineering team (15 developers) is still using it four years on. I'm a long-time TDD practitioner and I do a lot of code review, since catching a problem there is far cheaper than catching it in production. I've also built and maintained the CI/CD pipelines that get all of it shipped.
Technical Leadership & Mentorship	I've been the technical lead on small projects, most recently a 2-3 engineer rewrite of a legacy system, and earlier a 5-6 person team where I also ran Scrum. I set direction through RFCs, PR checklists, and API design guides, and I sit on AMA's Architecture Steering Committee. The mentoring side matters to me too: helping newer developers find their way through legacy code and getting a team comfortable with TDD.
Cloud-Native Infrastructure	I work across production AWS spanning multiple accounts (Lambda, EKS, SQS, RDS, IAM, ECR), with Terraform handling the infrastructure as code. I deployed a production EKS cluster running cert-manager, Karpenter, and Linkerd, and moved a brittle legacy document system onto it as containerized microservices with Helm. Most of what I build now leans on Postgres and Datadog for the data and the visibility.

Technologies

Backend Languages	Python, TypeScript, Java, C#, Node.js, Bash
Design & Practices	Object-Oriented Design, Design Patterns, TDD, Code Review, RFC Authorship
Architecture & APIs	RESTful APIs, OpenAPI, API Versioning, Microservices, Event-Driven Architecture
Databases	PostgreSQL, SQL, Aurora/RDS, Data Modelling, Redis
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM, ECR), Terraform, Kubernetes, Docker, Helm
CI/CD & Automation	CI/CD, GitHub Actions, Jenkins, GitOps, Infrastructure as Code
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, SLOs, Alerting